PAP – Many Daughters Project

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HP1: Having daughters leads to stronger support for gender equality (as measured by attitudes)

Variables:

"plj0620": Support for equal rights for women (N=164).

Available in years: 2016, 2017, 2018

Inclusion/exclusion criteria:

- We focus on **parents** (at least one son or daughter) **aged 25-55**.
- The period under investigation refers to data availability of the outcome variables (as summarized in Table above, column "years").

Exact analysis:

We consider the following model

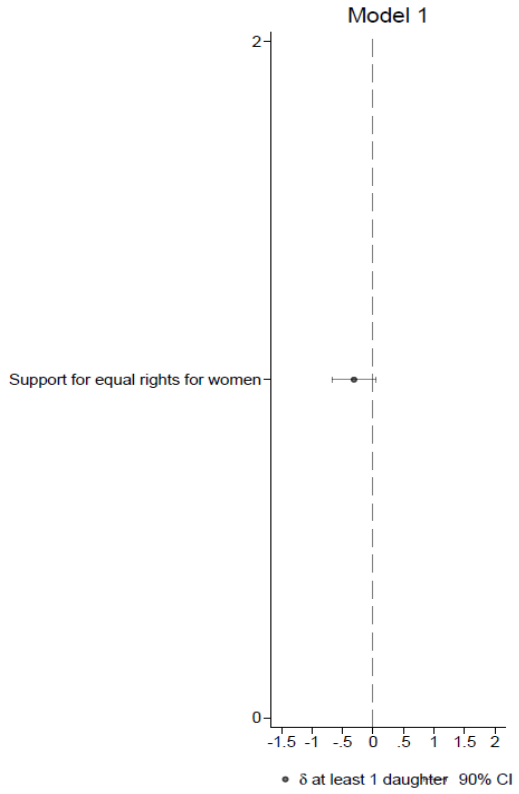
$$y_{it} = \delta * I[1 = N\ Daughters \geq 1] + \boldsymbol{X_{it}}\beta + \varepsilon_{it}$$

$\boldsymbol{X_{it}}$ includes the following **covariates**: respondents' age, respondents' gender, marital status (a dummy equal to 1 if married), origin (a dummy equal to 1 if foreign-born), education (three categories), household income bin (bottom 25% and top 25%), religion (four categories), year and state fixed effects.

The regression is estimated using robust standard errors and are weighted using the individual sampling weights (*phrf*).

No trimming or winsorization is applied to the data.

Results:



Interpretation:

The findings **show limited evidence in support of HP1** (i.e. having daughters leads to stronger support for gender equality measured in terms of "support for equal rights for women).

The estimated effect of having at least one daughter is small and statistically insignificant.

HP2: Having daughters leads to more left-leaning political preferences.

Variable

"plh0004": political tendency (left-right) (N=1993)

Available in years: 2005, 2009, 2014, 2019.

Given the nature of our hypothesis, our primary focus is on the political preference variables "PLH0004." To facilitate the interpretation of the coefficients, we made the following adjustment:

- We reversed the scale of PLH0004, so that a positive coefficient indicates stronger support for left-wing political orientations.

Inclusion/exclusion criteria:

- We focus on **parents** (at least one son or daughter) **aged 25-55**.
- The period under investigation refers to data availability of the outcome variables (as summarized in Table above)

Exact analysis:

As for HP1, we consider the model

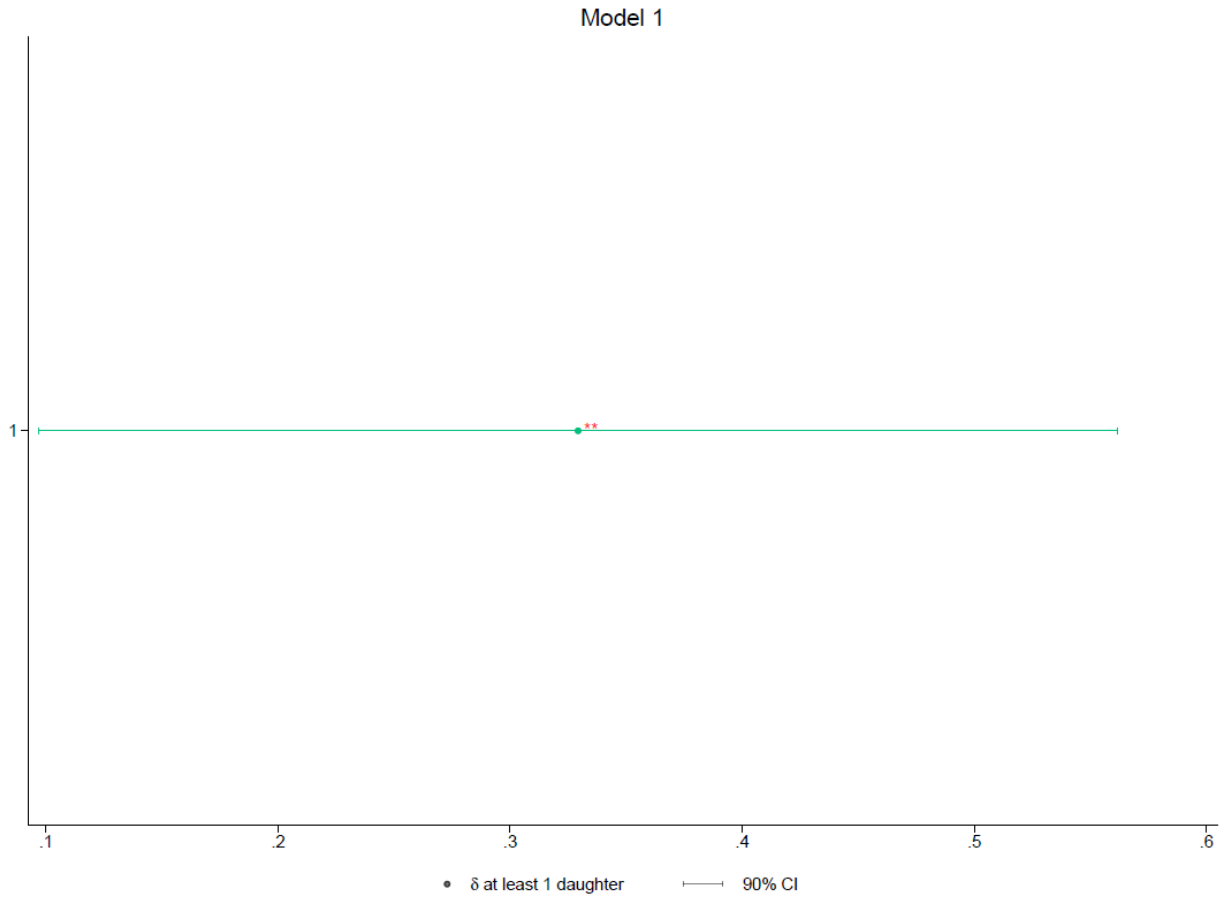
$$y_{it} = \delta * I[1 = N\ Daughters \geq 1] + \boldsymbol{X_{it}}\beta + 3_{it}$$

$\boldsymbol{X_{it}}$ includes the following covariates: respondents' age, gender, marital status (a dummy equal to 1 if married), origin (a dummy equal to 1 if foreign-born), education (three categories), household income bin (bottom 25% and top 25%), religion (four categories), and state fixed effects.

The regression is estimated using robust standard errors and are weighted using the individual sampling weights (phrf).

No trimming or winsorization is applied to the data.

Result:



Interpretation:

The results provide support for Hypothesis 2 (HP2), which posits that having daughters is associated with more progressive political attitudes.

Our model reveals a positive and statistically significant effect of having daughters on left-leaning political preferences.

HP3: Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

Variable:

"pli0044_v3": childcare activity during weekdays (N=6050)
Available from 2000 to 2019.

Inclusion/exclusion criteria:

- We restrict attention to households with **2 adults** (HH head and partner) **aged 25-55**, who **have children aged between 0 and 16**.
    - The period under investigation refers to the data availability of the outcome variables.

Exact analysis:

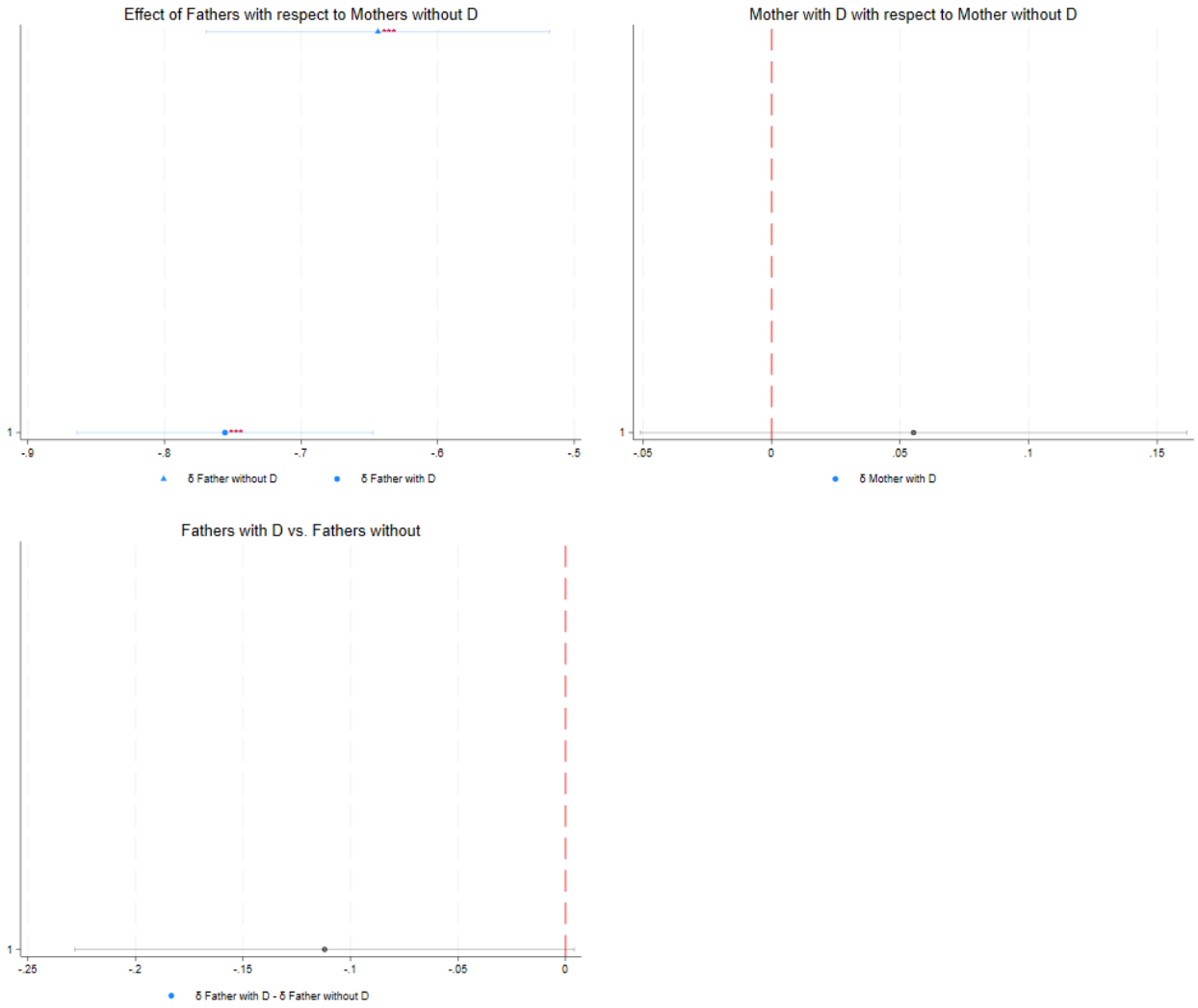
$$\log(y_{it}) = \delta^1 * I[1 = Father\ with\ at\ least\ one\ D] + \delta^2 * I[1 = Father\ with\ no\ D] + \delta^3 * I[1 = Mother\ with\ at\ least\ one\ D] + \boldsymbol{X_{it}}\beta + 3_{it}$$

$\boldsymbol{X_{it}}$ includes the following covariates: respondents' age, gender, marital status (a dummy equal to 1 if married), origin (a dummy equal to 1 if foreign-born), education (three categories), household income bin (bottom 25% and top 25%), religion (four categories), and state fixed effects.

The regression is estimated using robust standard errors and are weighted using the individual sampling weights (phrf).

No trimming or winsorization is applied to the data.

Results:



Interpretation:

The evidence we find offers limited support and appears (if anything) to contradict HP3.

The top-left graph displays estimates of δ^1 (circles) and δ^2 (triangles). The results confirm that fathers—both with and without daughters—spend less time than mothers on childcare.

We then compare fathers with daughters to those without. This comparison is more formally tested in the bottom-left graph, which reports the significance of the null hypothesis H_0: $\delta^1 - \delta^2 = 0$. If anything, fathers with daughters appear to spend slightly less time on childcare. These results provide evidence **counter to HP3.**

Eventually, the top-right panel presents estimates of δ^3. We find that mothers with daughters spend roughly the same amount of time on childcare and housework as mothers without daughters.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

Variables:

Same as for H1.

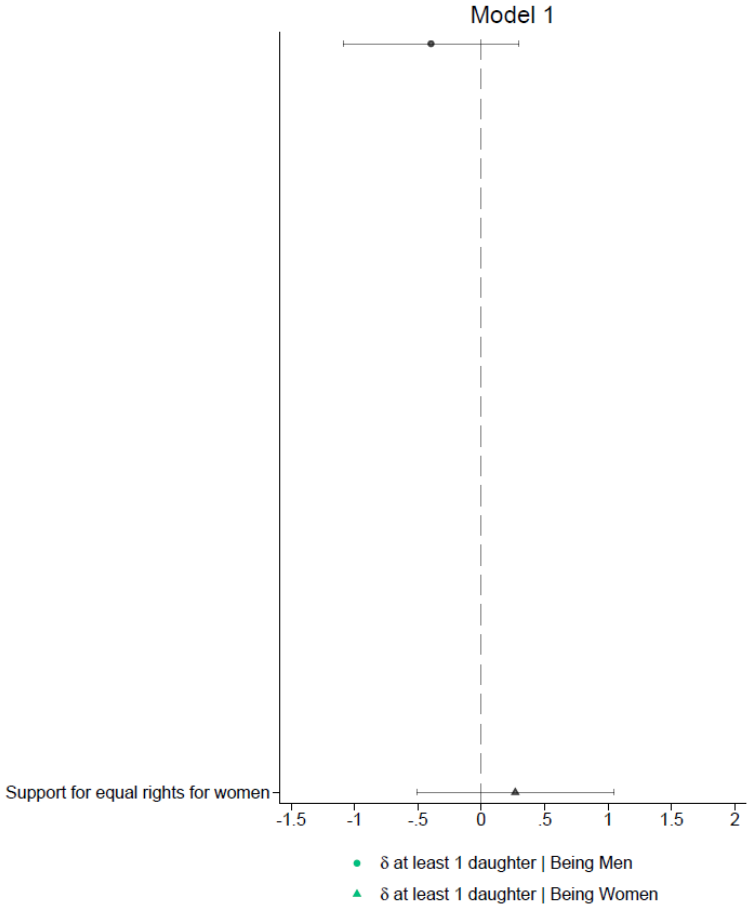
Inclusion/exclusion criteria:

Same as for H1.

Exact analysis:

Same as for H1, separately for fathers and for mothers. In the result tables are reported and compared both the coefficient and relative CI.

Results:



Interpretation:

The confidence intervals heavily overlap, indicating that the effect of having a daughter on the reported outcome (e.g. support for equal rights for women) does not differ significantly between fathers and mothers.

Possible future extensions (For all 4 hypotheses):

1) **Heterogeneity across parents' ages**: to explore whether the effect varies depending on how old the parent is.

2) **Dichotomizing the outcome (0/1) and applying a probit/logit model**: to test robustness using a binary specification.

3) **Using an ordered probit/logit model**: to account for the ordinal nature of the attitudinal outcome variables.

4) **Heterogeneity by the age of the daughter**: to explore potential mechanisms. If the effect is stronger for daughters in school years, it could suggest that exposure to gender equality messages through education and the resulting communication between daughters and parents drives the effect, rather than the mere experience of having a daughter.